

èfjç» - ç” µè`❖i¼š188-5665-3017 - é,®ç®±i¼šhs3434@foxmail.com -
ä, ºäºç½ ‘ç«™i¼šhs3434.github.io - **æ±
,è❖Ææ,,❖â❖ “**i¼šAMD ROCm Python â❖Žç«`â¼æâ❖ ‘â • ¥ç`â, ^i¼^ ROCm Radeon Cloud
æ-¹â❖ ‘i¼% --- ## ä, ºäºç®€ä» 3 â¹´ç” Ýç%©äzjæ❖`â • ¥ç`âÆ-ç»❖éâÆi¼Æè ± ºâ|è½-âž`
K8s â❖Žç«`æ ^â€, \*\*âº¹â™”âÆ-éf`ç½²â€❖èµ,,æº❖èºfâº|â€❖ä»»âŠiç¼-æž’ â€❖GPU
æž`ç❖ + æœ❖âŠi**â€, æœ€èz 2 â “è ± â| ROCm / K8s / Helm / FastAPI /
Prometheusi¼Æç«`ç«`âºÆæ^❖ [**k8s-llm-runtime**](https://github.com/hs3434/k8s-llm-
runtime)i¼^K8s ä, Šçš,, vLLM æ`jâž`æœ❖âŠiç½ ‘â...³i¼%oi¼Æ**v0.1.0
taggedi¼Æâ❖`çžºâœºè • ‘demo**â€, ç + Ýæ,% Snakemake ç±» Argo Workflows çš,, ML
â¹³â❖æš½è ± jâ€, --- ## æ±,è❖Ææš€èf½i¼^æ€%â²-â½❖âÆ¹é...❖æž’ âº❖i¼Æ4 â±
,i¼% ### â❖Žç«`æ , âzf - Python 3.11+i¼^asyncio, type hints, Pydantic, httpx asynci¼% -
FastAPIi¼^OpenAPI / dependency injection / lifespan i¼% - æµè` • i¼špytest, pytest-asyncio
âº¹â™”â , Žç¼-æž’ - Dockeri¼^multi-stage æž,,âºi¼ÆAlpine é • œâf❖â¼~âÆ-i¼% -
Kubernetesi¼^Pod / Service / Deployment / StatefulSet / ConfigMap / RBACi¼% - Helm
charti¼^values æ`jæ❖âÆ-i¼Æhelm install/upgrade/uninstalli¼ÆCI lint cleani¼% -
é • œâf❖â “âº “i¼šregistry.k8s.io / docker.io / ghcr.io / nvcr.ioi¼^â½â + ... DaoCloud
é • œâf❖âš é€Yi¼% ### GPU / LLM ç” Ýæ€❖ - vLLMi¼^OpenAI â...¼âº¹
APIi¼Ææž`ç❖ + æœ❖âŠi i¼% - AMD ROCm âºç;€i¼^HSA / HIP æ|,âzµi¼Æamd.com/gpu
èµ,,æº❖èºfâº|i¼% - PyTorch i¼^CNN æ— ¶ çººè Š £ç ❖â™” / Transformeri¼% - Hugging
Facei¼^transformers âº “i¼Ætokenizeri¼% ### â❖`è Š ,æµæ€ Š ä, Žâ • ¥â... é “¾ -
Prometheusi¼^prometheus_clienti¼Æhistogram / counter / gaugei¼% - structlogi¼^JSON
structured loggingi¼% - uv / ruff / mypy strict / GitHub Actions - Linux
æœ❖âŠiç«`èzç»`i¼^systemd, Nginx, ç«`âº£/èzç`çºç❖ + i¼% --- ## éiçºç»❖éâÆ ###
k8s-llm-runtimei¼šK8s ä, Šçš,, vLLM æ`jâž`æœ❖âŠiç½ ‘â...³i¼^v0.1.0 taggedi¼% **é—
ºèç`\*\*i¼^â-¹âº” JD è❖Æè`£ 1 + 3i¼%oi¼švLLM æ`jâž`éf`ç½²éœ€ GPU + âº¹â™” +
è • `ç” ±i¼Æâºçè`Yâºšâºâ...±âºæ— ¶ æ`jâž`çºç❖ + âºæ❖æ❖, i¼ç” çº ¶ /
âºâºçè`Yæf³ç” “ OpenAI â...¼âº¹ API æ— ¶ é—`æ Š é«~â€, i¼^ç»`æž`â-¹âº”
JD i¼š"âº Š æ`jâž`èç»f/æž`ç❖ + Workshopæ❖æ • ºç`â€❖âšYèf½æ¼” çºº”i¼%
æž ¶ æž,,i¼^â❖Æ chart + ä, %â±, lib + é • œâf❖â¼~âÆ-i¼%oi¼š - `charts/llm-inference/
`i¼švLLM è`Yè½½ charti¼Æ**GPU vendor â ± æ❖çâ...é❖ `values.yaml` **i¼^amd.com/
gpu / nvidia.com/gpu / none ä, %æ€❖i¼Ævalues æ~¾¼¼❖é€%æ©i¼% - `charts/llm-router/
`i¼šFastAPI ç½ ‘â...³ chart - ConfigMap æ³”â...¥ initContainer âš è½½â...¼Y chart - RBAC
æœ€âºæ❖fè™æi¼^Pod/Service/ConfigMap read+write + Leasei¼% - ServiceMonitor â❖`é€
%â❖`ç” “i¼^Service 0.0.0.0:9090/metricsi¼% - `src/k8s_llm_runtime/`i¼š3 â±, lib æš½è ±
j - `K8sJobOperator`i¼^â½Žâ±,i¼Æéšç” “ Job èºfâº|i¼% - `VLLMInferenceOperator`i¼^â ,
â±,i¼Æhelm CLI subprocess + GPU vendori¼% - `ModelOperator`i¼^é«~â±,i¼ÆOpenAI â...
¼âº¹ + è ± âš`éf`ç½²i¼% **4 âº Š æš€æœ-âºç,¹**i¼š 1. **lib ä, %â±,â + â±,æš½è ±
j**i¼^â-¹âº” JD â❖Žç«`æž ¶ æž,,èf½âš i¼% - ä, Šâ±, ModelOperator â❖â...³âzf
load_model(alias) / unload_model(alias) - ä, â±, VLLMInferenceOperator â❖è£... helm
ç» + èš, - â½Žâ±, K8sJobOperator éšç” “ K8s Job èºfâº| 2. **Helm ConfigMap æ³”â...¥
aliases**i¼^â-¹âº” JD âº¹â™”âÆ- + èµ,,æº❖èºfâº|i¼% - é❖â...❖æ`æ-jæ-ºâçžæ`jâž`éf½
rebuild é • œâf❖i¼^GPU é • œâf❖ 15-20GB rebuild 5-10mini¼% - ConfigMap
ç Š` çº Š ç” Ýæ • ^i¼Æé ¶ é • œâf❖é ± âºº 3. **K8s Lease â + â, fâ¼❖é” ❖**i¼^â-¹âº”
JD é + ç¾æèç»`ç»` + HAi¼% - coordination.k8s.io/v1 Lease é²æç Router
âºšâº-æœ-â¹ ¶ â❖ ‘éf`ç½²â❖Æä, €æ`jâž`- é” ❖ç²’ âº|æ€% model
aliasi¼Æè ± âš`èz ± æœY + ç»çº| 4. **â, šâŠiçº Š Prometheus metrics**i¼^â-¹âº” JD
â¹³â❖â❖`è Š ,æµæ€ Š i¼% - `chat_completions_duration_seconds`i¼^histogrami¼Æâ❖«

status code æ ð ç³⁄⁴i¹⁄⁴% - `http\_requests\_total`i¹⁄⁴^counteri¹⁄⁴Ææ% method/route/statusi¹⁄⁴%
- `models\_loaded`i¹⁄⁴^gaugei¹⁄⁴Æâððæ~ in-memory statei¹⁄⁴% - ServiceMonitor âð - é€%è¿
Grafana **GPU vendor â ð æð¢ demo**i¹⁄⁴^â¹âº JD âš â ð GPU é¿ ð ç³⁄⁴m¹⁄⁴%oi¹⁄⁴š -
values.yaml `gpu.vendor=amd|nvidia|none` - AMDi¹⁄⁴šamd.com/gpu èµ,,æºðè°fâºi¹⁄⁴ÆEROcm
vLLM é • œâfð - NVIDIAi¹⁄⁴šnvidia.com/gpu èµ,,æºðè°fâºi¹⁄⁴ÆCUDA vLLM é • œâfð -
nonei¹⁄⁴šCPU only æ · jâ¹⁄⁴ð¹⁄⁴^kind cluster é » èºm¹⁄⁴Ææœ-âœ°â¹⁄⁴€âð ‘ç’ ···i¹⁄⁴% **è
´··é ð æ¿ðè - ð**i¹⁄⁴š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint
â...` clean - 2 Helm chartsi¹⁄⁴^llm-inference + llm-routeri¹⁄⁴% **è , ©âð ‘ä , Žä¿ºâðð**i¹⁄⁴š -
Service DNS ä , ðâÆ¹é...ð bugi¹⁄⁴š`--set fullnameOverride={release\_name}` ä¿ºâðð - kind
èš,ç,¹â¹⁄²â ð ...é • œâfðâš é€Yi¹⁄⁴šcontainerdConfigPatches + DaoCloud mirror -
è¿¹²âºèf¹⁄²â ð ¹²â...¥ `docs/amd-interview-demo.md` ä³⁄⁴çŽºâœ°âð,è€f
æœ°æ¥¥i¹⁄⁴^v1.1i¹⁄⁴%oi¹⁄⁴šSSE æµðâ¹⁄⁴ðæŽ`çð ð / RBAC æ” ¶ ç´ s âðè - » / çœYâºŽ GPU
é¿ ð ç³⁄⁴e2e ### âYºâºŽ Snakemake çš,, ML
â¹³âððâ • ¥â¹⁄²œæµðæj ð æž ¶ i¹⁄⁴^ç” Yä¿¿èfÆæ¹¹⁄⁴Æç± » Argo Workflowsi¹⁄⁴% -
èfÆæ¹¹⁄⁴šSnakemake rule è - æ³ • æ— æ³ • âºžçŽºæ · jâð—ésðæ,,ðç,,âð^ -
èº³⁄⁴èºi¹⁄⁴šæ ‘`â¹⁄⁴f rule è - æ³ • i¹⁄⁴Æç´ æŽ¥è°fâ ð ...éf`æŽ¥âð£âš`æ€ðç” Yæ^ð
rulei¹⁄⁴Æè s ,èÆfè³⁄⁴ “â...¥è³⁄⁴ “â ð °ç± »âžçâð³âð`ç s - æœ · â¹⁄⁴ðæ¹⁄⁴æŽ¥ - â...³èð” i¹⁄⁴šJD
âš â ð ð é¿¹²æœ°â¹¹`â¿¹ â¹³âððç” âð ‘ç»ðé°Æ¹â€” â€” æœ-é¿¹çºæ · ML â¹³âððæš¹⁄²è±
jçš,,â • ¥ç`âºžè • µ ### è,, ‘æœ°æŽ¥âð£ CNN è s £ç ðâ¹¹⁄⁴^PyTorch¹⁄⁴%â€” â€” 6 â ´`â -
|ä¹ è • `çº¿âº s â ð ° - ç«`â^°ç«`æ— ¶ çº°âð • ç s - è s £ç ð EEG ä¿jâð • - PyQt âºžæ— ¶ GUI +
FastAPI æš´éœ² - **ä” “âº “**i¹⁄⁴š[github.com/hs3434/bci-pipeline-demo](https://github.com/
hs3434/bci-pipeline-demo) --- ## â • ¥â¹⁄²œç»ððŽ ð ### ä , Šæµ • âº
mè ð Æâð¿jæ¿ð`ç s ‘æš€æœ%é¹²ðâ...-âð , i¹⁄²œ ç” Yç%©ä¿jæð`â • ¥ç`â , ^ i¹⁄²œ
2025.04 - è ð³âš - **Docker éf`ç¹⁄²**i¹⁄⁴šä , šâš¿æ£ç ð Docker âÆ-i¹⁄⁴^âYºâºŽ snakemake +
è ð³ç” æj ð æž ¶ i¹⁄⁴% - **æœððâš¿ç`è¿ðç»`**i¹⁄⁴šLinux æœððâš¿æ¹¹⁄⁴æ—
¥â , ç»`æšm¹⁄⁴ÆNginx âðððâð ‘ä»£çð ð i¹⁄⁴Æsystemd æœððâš¿çº¿çð ð -
âº ‘â¹³âððâ • ¥â... • â¹⁄⁴€âð “i¹⁄⁴šä , šâš¿è ð³âš`âÆ-æµðç`ç + çº—æ³ • â • ¥â... • ###
ä , Šæµ • æ- s æ~ “ç” Yç%©âÆ»â¿ç s ‘æš€æœ%é¹²ðâ...-âð , i¹⁄²œ
ç” Yç%©ä¿jæð`ç” âð ‘â • ¥ç`â , ^ i¹⁄²œ 2022.08 - 2024.07 - **âº ‘â¹³âððç” Yä¿¿â • ¥â...
• â¹⁄⁴€âð “**i¹⁄⁴šR / Python / Linux / Docker / âº ‘âžYç” Y - **ä , šâš¿è ð³âš`âÆ-**i¹⁄⁴šâÆ»â -
|è¹⁄²-â¹⁄² • ç»,,ä , šâš¿æµðç`ç`è ð³âš`âÆ-i¹⁄⁴Ææ`jâð—âÆ-â€ððâð`âððç” · -
â¹³âððâœ°âð€i¹⁄⁴š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ###
ä , Šæµ • æ- s æ~ “ç” Yç%©âÆ»â¿æœ%é¹²ðâ...-âð , i¹⁄²œ ç” Yç%©ä¿jæð`âºžâ¹ ç” Y i¹⁄²œ
2021.10 - 2022.01 - âð • ç» ð èfžè¹⁄²-â¹⁄² • ç»,,i¹⁄⁴šè
´`æž s ä€ðâ , è s ,â ð ð æžðæ€ðâ” âððžæ€ðä , æ€ s âÆ-â ð ð æžð -
ä , »èðæš€æœ¹¹⁄⁴šâð • ç» ð èfžè¹⁄²-â¹⁄² • ç»,,â€ðRâ€ðPythonâ€ðLinux --- ##
æ • ¹²è²èfÆæ¹¹⁄⁴æ—â ð ð ææž—ç s ‘æš€â s â¿¹¹⁄²œ æðç%©ç s ‘â -
|ä , Žæš€æœ-ä , “ä , š i¹⁄²œ æœ-ç s ‘i¹⁄²œ 2018.09 - 2022.07 i¹⁄²œ GPAi¹⁄⁴š3.35 - æ - ä , šèºæ-
ð i¹⁄⁴šâ€šâYºâºŽé€»è³⁄⁴ ‘æ-`è’ ,âžâ¹⁄² ç¿âºšâºéº|â€” ä¹⁄⁴ç , â...³æ€ s çš ¶ âðšæš—
â€” æ€ s çš ¶ æÆ ð æ ð ð æ¿ - â s â`é¿¹çºi¹⁄⁴šæ¹çºmèðÆæŽ¥ç s ðâ¹ç´«èš±èœè “¿æš—
é “œæ±jæY “çš,,â¹⁄²±â “ð¹⁄⁴^âð ,ä , Žâ¹⁄²œè€...i¹⁄⁴% --- ## ä , äººâ¹⁄⁴^âš¿ - **6 â ´`è ð³â¿ K8s
âðžç«`æ ^\*\*i¹⁄⁴šâ»žé» ¶ â^º v0.1.0 tagged é¿¹çºi¹⁄⁴Æç«`âºÆæ^ððÆ Helm chart + Python
lib - **çŽºâœ° demo èf¹⁄²âš»**i¹⁄⁴šâð`è ð³â , |ç” µè,, ‘âœ` kind cluster ä , Šè • ‘é€šâºÆæ • ´
LLM æŽ`çð ð æµðç`ç`i¹⁄⁴^é¿¹ç¹⁄² â ð ð è • `ç” ± â ð ð chat completions â ð ð metricsi¹⁄⁴% -
â¿¹ èf¹⁄²âš»â¹⁄⁴ºi¹⁄⁴š3 â¹`â ð ð ...ä»žç” Yç%©ä¿jæð`è • `ç • Æâ^º K8s
âðžç«`i¹⁄⁴Ææš€æœ`çfæf...é«` - \*\*3 â¹` Python â • ¥ç`âÆ-
ç»ðé°Æ¹¹⁄⁴šâ»žç s ‘ç” è,,šæœ-â^ºâº ‘â¹³âððâ • ¥â... • çš,,âºÆæ • ´è • -â³⁄⁴,,